OHDSI Study Agent for Human–AI Team-Based

Real-World Evidence Generation in the Veterans Health Administration

Richard D. Boyce, PhD | University of Pittsburgh | VA Informatics and Computing Infrastructure (VINCI)

HYPOTHETICAL SYMPOSIUM POSTER • ARCHITECTURE / FUNCTIONAL-PROTOTYPE REPORT • SCIENTIFIC REVIEW REQUIRED

CENTRAL TAKEAWAY

Reusable, governed AI support for OHDSI workflows combines ACP/MCP coordination with deterministic validation — while researchers remain accountable for consequential decisions.

WHY GOVERNED TEAMING?

RWE study design links judgments about phenotypes, causal assumptions, cohorts, and interpretation. AI assistance is bounded where researchers need structured support.

RESEARCHER OVERSIGHT

- Explicit confirmations and stage gates
- Validated outputs and auditable routing
- Sanitized summaries / metadata to models
- Fail-closed PHI/PII sanitation where applicable
- Row-level clinical data are not sent to the model

PROTOTYPE MATURITY

Early functional prototype: approx. TRL 4–5 (scripted laboratory demonstration).

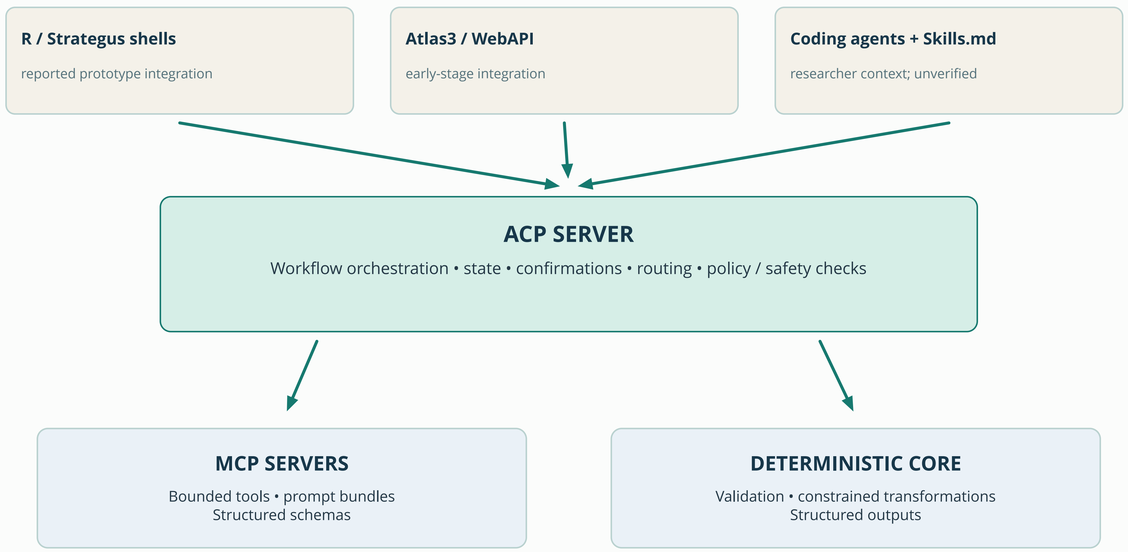
Not established: efficiency, decision accuracy, calibrated reliance, or better study outcomes.

SCOPE OF CLAIM

The generative model is coordinated by an agent harness; Study Agent does not bundle a model and does not replace researcher judgment.

1 SERVICE-BASED, GOVERNED ARCHITECTURE

Distinct roles: clients access shared services; ACP, MCP and deterministic code are not interchangeable.



Source status: R/Strategus demonstrated; Atlas3/WebAPI early-stage. Coding-agent support and exact OHDSI deterministic package bindings are intentionally not represented as verified implementation.

3 WORKFLOW-CONTEXTUAL GUIDANCE VIA /ohdsi

EXAMPLE QUESTION

“How should I think about time at risk in this incidence design?”

WHAT /ohdsi ADDS

The shell supplies workflow type, current step, active role, selected cohorts/artifacts and constraints. ACP returns a normalized response with step-specific guidance, cautions and next actions.

The poster shows the contextual mechanism, not a fabricated literal LLM reply.

2 NAMED GOVERNED FLOWS

Excerpt of Table 1 — stable endpoints with defined inputs, outputs and validation.

phenotype_recommendation
phenotype_make_computable
concept_sets_review
cohort_critique_general_design
workflow_context_dialogue

Flows apply deterministic checks at each stage; restricted outputs identify user-confirmable steps before consequential writes.

APPLICATION EVIDENCE

NO APPROVED AUTHENTIC SCREENSHOT AVAILABLE FOR THIS PUBLIC POSTER BUILD

Placeholder only — not a system interface or scientific evidence. Replace only with an approved demo/repository artifact after source review.

INTERPRETATION, LIMITS & PROVENANCE

Architecture and functional-prototype report; not evidence of improved efficiency, decision accuracy, clinical safety or study outcomes.
Scientific source: supplied private manuscript. Design decisions: DESIGN_BRIEF.md. No patient-level data, credentials or authentic screenshot assets are included.

Selected sources in supplied manuscript: Study Agent repository; SlashOhdsiStrategusAssistant; SlashOhdsiAcpClient; Study Agent / Atlas3-WebAPI demonstrations. Verify URLs, branding and submission metadata before any public release.

EDITABLE SVG • 48 × 36 in • build.py + poster.toml
No QR destination supplied; none generated.